

Nicole Timko

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EDUCATION

Carnegie Mellon University, Dietrich College of Humanities and Social Science

Pittsburgh, PA

BS in Statistics and Machine Learning | **GPA: 3.87**

May 2027

Coursework: Principles of Imperative Computation (C), Modern Regression, Probability and Statistics Inference I/ II, Concepts of Mathematics, Multivariate Analysis, Matrix Algebra, Advanced Methods of Data Analysis, Algorithms and Advanced Data Structures

Christopher Newport University (transferred after sophomore year)

Newport News, VA

Majored in Computer Science | Minor in Leadership before transferring | **GPA: 4.0**

2022-2024

NCAA ATHLETICS AND LEADERSHIP

Carnegie Mellon University Women's Varsity Basketball | *Varsity Athlete*

2024-2027

- Selected by coaching staff and teammates as team captain after taking on an expanded leadership role, serving as a primary link between coaches and players to improve communication, accountability, and team culture
- Fostered an inclusive team environment by mentoring underclassmen, supporting teammates through challenges, mediating conflicts, and ensuring all players felt heard and valued
- Provided in-game leadership by identifying key areas of focus, communicating tactical adjustments, and reinforcing team standards through consistent work ethic and preparation

Christopher Newport University Women's Varsity Basketball | *Varsity Athlete*

2022-2024

WORK AND RESEARCH EXPERIENCE

Machine Learning Internship - BTASports

June 2026-August 2026

- Built a leakage-safe, 3-stage NBA prediction pipeline to predict player minutes, 11 player-prop outcomes, and game-level results using expanding window walk forward validation and evaluating different models using calibration and error metrics.
- Developed a K-Means clustering pipeline to segment bettors into 5 behavioral archetypes, using standardized features and selecting the optimal number of clusters through the elbow method and silhouette analysis.
- Built a mixed-integer linear programming optimizer to generate optimal NFL daily fantasy lineups under salary cap, roster, and positional constraints across two contest formats; validated solutions using engineered test cases.

FTN's API Project - Pittsburgh, PA

June 2026-July 2026

- Engineered a NBA dataset from FTN's API, parsing nested JSON across multiple endpoints and validating cross file consistency to identify an upstream data inconsistency that was reported to the API owner for correction.

Tartan Athletics Analytics Project - Database Development | Pittsburgh, PA

January 2026-May 2026

- Architected an athletic performance tracking system using R Shiny and PostgreSQL database to centralize testing data across teams while designing interactive dashboards with dynamic visualizations for performance analysis.
- Engineered an automated pipeline to standardize CSV uploads, transform wide-format testing data into a long-format SQL schema, and detect misspelled player names using Jaro Winkler string distance.
- Implemented database connectivity via DBI and Postgres, to enable real-time create, read, update, and delete operations and automatically refresh summary statistics upon data interaction.

Summer Undergraduate Research Apprenticeship - Basketball Analytics | Pittsburgh, PA

May 2025-August 2025

- Developed a reusable Python web scraping script to programmatically extract roster and play-by-play data from the CMU Athletics Website, adaptable for past and future seasons.
- Built a reproducible R pipeline to clean, aggregate, and analyze lineup-level data, applying ECDF-based thresholds and nonparametric tests to evaluate performance differences between 3-guard and 4-guard lineups.

SKILLS

Languages & Technical: Python, R, SQL, Java, C, Statistical Analysis, Optimization, Data Engineering, Data Visualization, Web Scraping, GitHub, Database Development, Problem Solving, Debugging

Leadership & Interpersonal: Team Leadership, Communication, Mentorship, Collaboration, Conflict Resolution

AWARDS AND HONORS

Deans List | Carnegie Mellon Dietrich College (4 semesters, 2025-2026), Christopher Newport (4 semesters, 2022-2024)

All UAA First Team · All Region 7 2nd Team · UAA All Academic Team

2026

Poster Presentation at Carnegie Mellon Sports Analytics Conference | Pittsburgh, PA

October 2025

College Sports Communicators Academic All-District Team

2024, 2025

Canon Scholarship for Engineering and STEM (Computer Science - Christopher Newport University)

2 years